Predicate Assignment Questions

1. Odd Number Checker

Write a `Predicate<Integer>` that checks if a number is odd.
Task: Test it on numbers from 1 to 10 and print only odd ones.

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2. Non-empty String

Create a `Predicate<String>` that checks if a given string is not empty and not null.
Task: Filter valid strings from a list: `["", "Java", null, " "]`

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3. Student Pass Filter

Given a `Student` class with `name` and `marks`, use a `Predicate<Student>` to check if marks are ≥ 40.
Task: Print only passing students from a list of 5.

◆ Supplier Assignment Questions

1. Random Number Generator

Use a `Supplier<Double>` to generate and print 5 random numbers between 0 and 1.

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2. Default City Provider

Create a `Supplier<String>` that supplies a default city name (`"Pune"`).
Task: Use it when no user input is provided (simulate fallback).

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3. Product List Generator

Create a `Supplier<List<String>>` that returns a list of 5 default product names.
Task: Print each product using a loop.

◆ Function Assignment Questions

1. String Length Calculator

Write a `Function<String, Integer>` to return the length of a given string.
Task: Apply it to 3 names and print results.

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2. Temperature Converter

Create a `Function<Double, Double>` to convert Celsius to Fahrenheit.
Formula: `F = (C × 9/5) + 32`
Task: Convert and print values for 0°C, 20°C, 37°C.

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3. Student Grade Generator

Given a `Function<Integer, String>` that returns grade ("A", "B", "Fail") based on marks:

- ≥ 75 = "A",
- 50–74 = "B",
- below 50 = "Fail"
  Task: Use it to grade a list of 5 students' marks.

◆ Consumer Assignment Questions

1. Print String in Uppercase

Write a `Consumer<String>` to print a given string in uppercase.

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2. List Printer

Create a `Consumer<List<String>>` that prints each element in a list using a loop.

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3. Invoice Formatter

Create a class `Invoice` with fields `itemName`, `quantity`, and `price`.
Write a `Consumer<Invoice>` to print a formatted bill line:

Example: `Item: Pen | Quantity: 2 | Total: ₹20`
Task: Use it for 3 invoice objects.